

Welcome to Java

INTRODUCTION TO JAVA



Jim White
Java Developer

Your instructor - Jim White

- Software developer and manager
- Working with Java since (almost) its inception



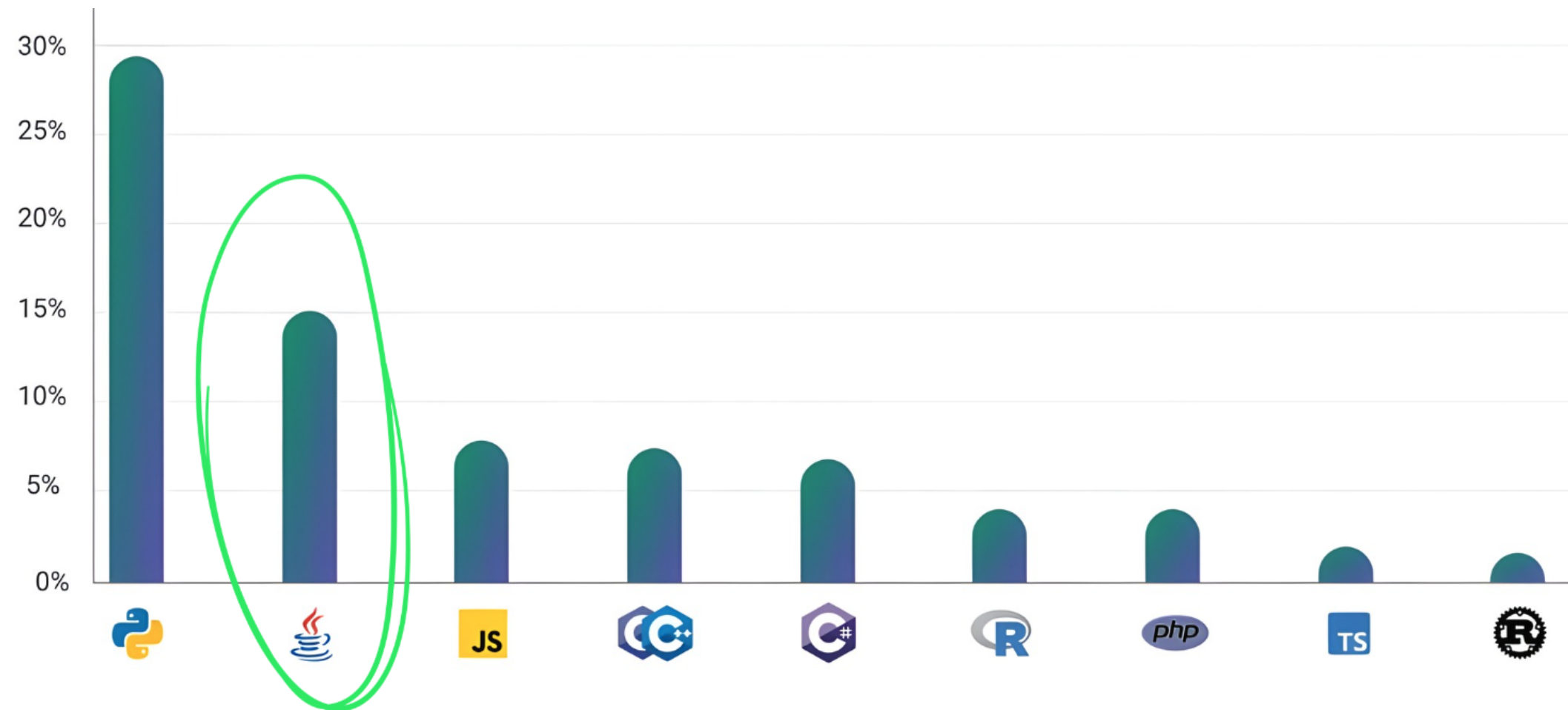
Who is this course for?

- Anyone
- Java:
 - Syntax is stricter and less intuitive than, e.g., Python or JavaScript
 - Flexible and scalable
- Steeper learning curve



What is Java?

- High-level, general-purpose, memory-safe, object-oriented programming language



Popularity of Programming Languages

¹ <https://softjournal.com/insights/is-java-still-used>

Where is Java?

Google

amazon

NETFLIX



android

...

Starting out

```
class WelcomeToJava {  
  
    public static void main(String[] args) {  
  
    }  
  
}
```

- Code is organized into classes
- Method `public static void main(String[] args)`
 - entry point of our program

Printing out - how not to do it

```
class WelcomeToJava {  
  
    public static void main(String[] args) {  
        "Hello Java!"  
    }  
  
}
```

```
src/Main.java:8: error: not a statement "Hello Java!"  
src/Main.java:8: error: ';' expected "Hello Java!"  
2 errors error: compilation failed
```


Printing out - the correct way

```
class WelcomeToJava {  
  
    public static void main(String[] args) {  
        System.out.println("Hello Java!")  
    }  
  
}
```

Hello Java!

- Printing requires double quotes `" "`

Let's practice!

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Interpreting Java syntax

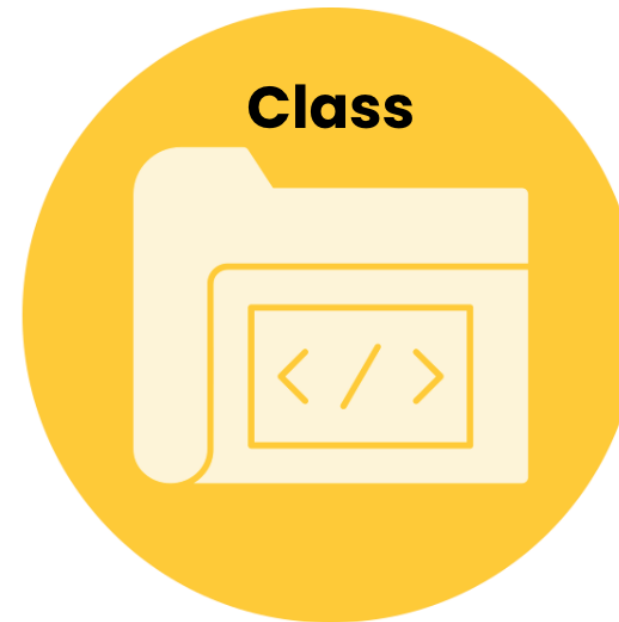
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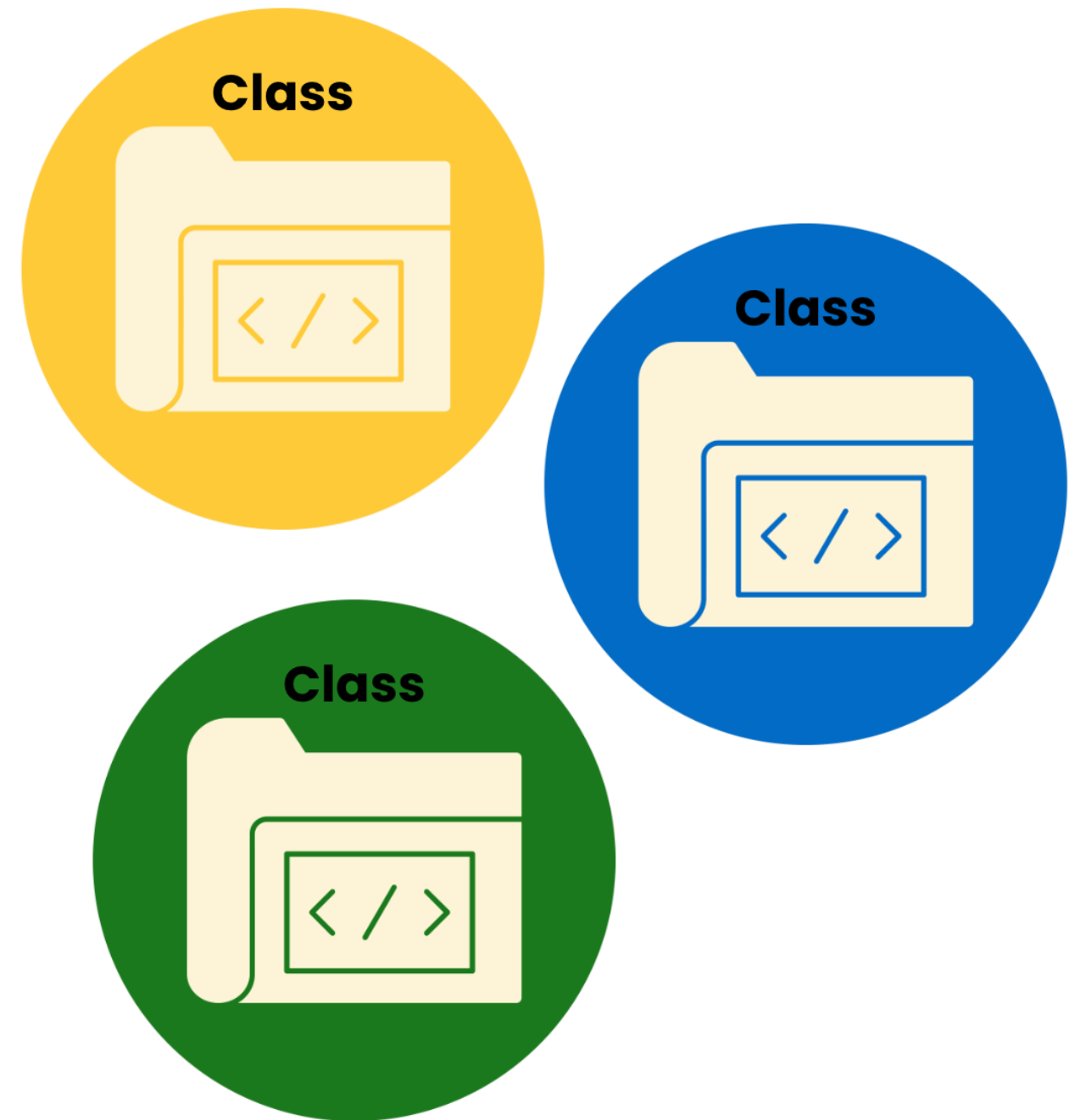
Class

- A container to organize code



Class

- A container to organize code



Defining class

```
class Greeting {
```

- `{ }` mark where class begins and ends

Class and the main method

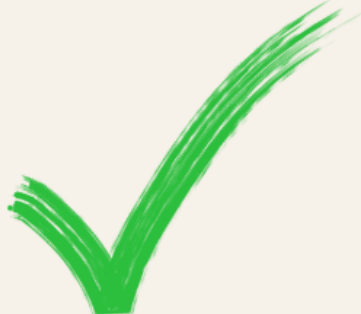
```
class Greeting {  
  
    public static void main(String[] args) {  
  
    }  
  
}
```

- Code runs inside the method `main()`

Case, semicolons, and white space

- Java is case sensitive
 - `main` $\neq$ `Main`
- Every line inside the method ends with `;`
 - `System.out.println("Hello!");`
- Indentation is only used to improve readability

```
class WelcomeToJava {  
    public static void main(String[] args) {  
  
        System.out.println("Hello!");  
  
    }  
}
```



```
class WelcomeToJava {  
  
    public static void main(String[] args) {  
  
        System.out.println("Hello!");  
  
    }  
}
```



Println as a calculator

```
class PrintlnAsCalculator {  
    public static void main(String[] args) {  
  
        System.out.println(1+2);  
    }  
}
```

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Adding comments

```
class PrintLnAsCalculator {  
    public static void main(String[] args) {  
        // We can use println as a calculator  
        System.out.println(1+2);  
    }  
}
```

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- Comments make code easier to read

Let's practice!

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Starting with variables

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Variables are boxes



Variables are boxes



- Allow to store different kinds of information
- Have a type, a name, and a value

Assigning variables - String keyword

String

Components:

- String keyword

Assigning variables - name

```
String variableName
```

Components:

- `String` keyword
- Variable name

Assigning variables - syntax

```
String variableName =
```

Components:

- `String` keyword
- Variable name
- `=` to assign value

Assigning variables - text

```
String variableName = "text in double quotation marks";
```

Components:

- `String` keyword
- Variable name
- `=` to assign value
- The actual value
- Semicolon `;`

```
String message = "Hello, Java!";
```

Assigning variables - whole numbers

- Use `int` for whole numbers

```
int age = 29;
```

Single equal sign

- Doesn't mean mathematical equal

=

≠

- Assignment operator



Declaring variables

- Always specify the data type
 - Text -> `String`
 - Number -> `int`
- Other data types: `char` , `boolean` , `float` , and more



Naming convention

- Good variable names:

```
int age = 31;

String welcomeMessage = "Hello";

String daysToSummer = "Too many";

String myName = "Jim";
```

- Bad variable names:

```
int 4age = 31;

String welcome message = "Hello";

String days_to_summer = "Too many";

String MyName = "Jim";
```

- Use camelCase: lowercase first word, capitalize next words

Using variables

```
class InfoAboutJava {  
    public static void main(String[] args) {  
  
        String message = "Java is awesome!";  
  
        // Print the message variable  
        System.out.println(message);  
    }  
}
```

Java is awesome!

Summary

- Variables store values
- `type variableName = value;`
- Variable names:
 - Start with letter
 - Follow camelCase
 - Cannot contain space

- Use variables instead of their value

```
class InfoAboutJava {  
    public static void main(String[] args) {  
  
        String message = "Java is awesome!";  
  
        // Using variable directly  
        System.out.println(message);  
    }  
}
```


Let's practice!

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